

QUORIN TECH · FOR MACPROAUTOMATION

MachineIQ ERP Release Plan

Revised delivery sequence · 07 September 2026

The Machine Project is the central business record.

Enquiry → Quotation → Sales Order → Machine Project → Engineering → BOM → Availability → Purchase → Assembly
→ Quality → FAT → Delivery → SAT / Commissioning → Invoice → Warranty → Service

Release sequence

Release	Name	Business outcome
R1	Foundation & Master Data	Trusted organization, identity and reusable business masters.
R2	Sales & Machine Project Initiation	Accepted demand becomes an approved order and a traceable machine project.
R3	Project Planning & Engineering	Engineering responsibilities, dates, documents and changes are controlled.
R4	BOM & Material Planning	Progressive engineering BOM becomes controlled material demand.
R5	Inventory & Warehousing	Available, reserved and moving stock is visible by project.
R6	Purchase & Supplier Management	Project shortages are procured and supplier commitments are controlled.
R7	Production, Assembly & Quality	Machines are built with controlled material, inspection and rework records.
R8	FAT, Delivery & Commissioning	Machines are accepted, delivered, installed and formally handed over.
R9	Finance, Workforce & After-Sales	Project cost, cash, people and installed-machine service history are connected.
R10	Enterprise Integration & Intelligence	The platform gains enterprise workflows, integration and cross-module intelligence.

Source hierarchy. [Dashboard.docx](#) is the original customer-provided requirements document. [Product Specification v1.0](#) (07 September 2026) defines the revised workflows, release boundaries and acceptance criteria. This plan summarizes that specification; it does not replace the original requirements.

Plan revision 2 · Replaces the earlier inventory-first release sequence. R1–R10 are product release identifiers, not software version numbers or delivery dates. This document is a plan, not a completion certificate.

RELEASE SCOPE & ACCEPTANCE**R1****Foundation & Master Data**

Trusted organization, identity and reusable business masters.

- Company profile, legal/tax details, branches, physical locations, departments, currency, timezone and fiscal settings.
- Users, employee identity, roles, permissions, credentials and access assignments; active/inactive controls.
- Customer and supplier codes, addresses/sites, contacts, tax data, currencies, payment terms and supplier lead times.
- Items, categories, UOM, make/buy, cost, selling price, manufacturer/part number, stock flag and reorder level.
- Warehouse master, configurable tax/VAT, currency, reference statuses, document types and numbering series.

Reports · Master completeness, active users/roles, customer and supplier lists, item catalogue and change history.

Acceptance · Administrators configure hierarchy/access without code changes; duplicate codes and invalid mandatory data are rejected; privileged changes are audited; downstream modules select consistent masters.

Sequence / boundary · Foundation for every release. Workforce operations expand in R9; stock transactions start in R5. Spec §6, p.8.

R2**Sales & Machine Project Initiation**

Accepted demand becomes an approved order and a traceable machine project.

- Enquiry: machine/application, quantity, target delivery, customer site, source and attachments; requirements for use, capacity, interfaces, utilities, constraints, standards and customer-provided items.
- Technical-commercial quotation: scope, exclusions, price/tax, validity, delivery, warranty and payment milestones.
- Quotation revisions/comparison, internal approval, customer acceptance/rejection and conversion history.
- Sales order: customer PO, ordered scope/value, delivery, payment milestones and special terms.
- Project creation from accepted order: project code/owner, machine category, site, dates, value and baseline; communications, comments, attachments, activity and status dashboard.

Reports · Sales pipeline, quotation conversion, quotations by customer, orders by month/customer, pending orders, payment milestone schedule and project intake.

Acceptance · Accepted quote creates an order without re-entry; approved order creates a project with source references; every revision remains visible and only the current approved version is operational; required references, dates and values are validated.

Sequence / boundary · Uses R1 masters. Basic approvals ship here. Delivery execution is R8; invoices and payment allocation are R9. Spec §7, p.9.

RELEASE SCOPE & ACCEPTANCE

R3

Project Planning & Engineering

Engineering responsibilities, dates, documents and changes are controlled.

- Project templates, phases, milestones, stage gates, planned dates and responsible departments.
- Mechanical, electrical, controls/software and other work packages; engineer/team allocation, task ownership, dependencies, effort, dates and completion evidence.
- Requirement register with source, owner, priority, verification method, status and evidence links.
- Document/drawing register, revision, review, approval, release and superseded history; design-review actions and decisions.
- Risk, issue and action registers; engineering changes with affected records, cost/schedule impact, approval and implementation tracking.

Reports · Timeline/project health, milestone performance, overdue tasks, workload, document status, requirement coverage, risk exposure and change impact.

Acceptance · Managers assign dependent work; released documents require new revisions for change; change history identifies affected records; progress derives from controlled work.

Sequence / boundary · Builds on the R2 project baseline; provides engineering ownership and release control for R4. Spec §8, p.10.

R4

BOM & Material Planning

Progressive engineering BOM becomes controlled material demand.

- Preliminary, engineering and manufacturing BOMs; multi-level assemblies, configurable sections and engineer/work-package ownership.
- Lines: item, quantity/UOM, make/buy, required date, preferred supplier, manufacturer part, cost and notes; validated new-item creation during authoring.
- Alternatives/substitutions, customer-supplied and non-stock items, scrap allowance and long-lead identification.
- Revision comparison, line-change history, approval and partial section release before the full assembly BOM is complete.
- Availability, reservations, on-order quantities, shortages and required-by dates; estimated cost versus budget; CSV/Excel import/export and CAD/BOM import preparation.

Reports · BOM cost, revision variance, shortages, long-lead items, make/buy, unreleased sections and material readiness.

Acceptance · Designers contribute independently; only released quantities drive demand; revisions show added/removed/changed lines; demand retains project, machine, BOM revision and required date.

Sequence / boundary · Uses R1–R3. Availability/reservation and purchase views connect progressively as R5/R6 introduce live stock and purchasing. Spec §9, p.11.

RELEASE SCOPE & ACCEPTANCE

R5

Inventory & Warehousing

Available, reserved and moving stock is visible by project.

- Warehouse, zone/bin or configurable storage locations; opening balance, stock in/out, transfer, adjustment and cycle count.
- Stock ledger and balances by item/location, with batch/lot and serial tracking where enabled.
- Project reservation, allocation/deallocation, material issue and unused-material return.
- Available-to-promise separates on-hand, reserved, available, inspection and blocked stock; minimum stock, reorder proposals and alerts.
- Negative-stock policy, controlled adjustment approval, organization valuation method and complete movement history.

Reports · Current stock, reservations, movements, valuation, ABC analysis, slow-moving items, shortages, cycle-count variance and reorder list.

Acceptance · Posting changes balances atomically; posted records cannot be edited; reversal creates linked counter-transactions; stock conditions remain distinct; issues cannot exceed permitted availability/reservation.

Sequence / boundary · Uses R1 masters and R4 released demand. Connects inventory to R6 receipts and R7 production. Spec §10, p.12.

R6

Purchase & Supplier Management

Project shortages are procured and supplier commitments are controlled.

- Purchase requests from released BOM shortages or manual entry, with requester, project, need date and justification; module approval.
- RFQs to suppliers; quotation capture for price, tax, lead time, validity, freight, terms, deviations and documents.
- Technical/commercial comparison and supplier selection with justification.
- PO with project/BOM references, delivery schedule, terms, approval and dispatch status; amendments/history and controlled outstanding-quantity closure.
- Partial GRN, inspection/accepted/rejected quantities, purchase returns and pending balances; supplier invoice matching, payment status and AP handoff.

Reports · Open POs, pending/delayed deliveries, purchase/price history, supplier performance, pending GRN/inspection and project purchase cost.

Acceptance · Released shortages create requests without re-entry; selection records its reason; partial receipts update pending quantity/inspection stock; PO amendments retain prior versions and reapproval.

Sequence / boundary · Uses R4/R5. Quality inspection integrates with R7; supplier payment processing and allocation are R9. Spec §11, p.13.

RELEASE SCOPE & ACCEPTANCE

R7

Production, Assembly & Quality

Machines are built with controlled material, inspection and rework records.

- Production planning by project/machine and work centre/team; work orders for mechanical, electrical, controls/software and configurable operations.
- Material issue, actual consumption, returns, substitution and variance; operation ownership, planned/actual effort, blockers and completion evidence.
- Production output/finished-machine serial number and as-built BOM reference.
- Reusable inspection plans, characteristics, tolerances, evidence and disposition; incoming, in-process and final inspection linked to GRN/work order/output.
- NCR/CAPA: containment, root cause, corrective/preventive actions, verification and closure; rework and approved deviation/concession.

Reports · Work-order status, WIP, consumption/variance, productivity, readiness, inspection results, NCR ageing, rework and supplier quality.

Acceptance · Blocking NCRs/mandatory inspections prevent completion; consumption/output update inventory and cost sources; failed characteristics receive disposition; serialised output retains BOM and quality traceability.

Sequence / boundary · Uses released BOM, stock and purchasing from R4–R6; supplies readiness evidence for R8. Spec §12, p.14.

R8

FAT, Delivery & Commissioning

Machines are accepted, delivered, installed and formally handed over.

- FAT plan, checklist/protocol, prerequisites, participants, dates and test evidence; deviations and customer observations.
- Punch items with severity, owner/due date and closure; FAT approval/sign-off and dispatch controls.
- Packing list, delivery note, partial delivery, shipment/dispatch, carrier and delivery confirmation.
- Installation/site-readiness plan, engineers and site activities; SAT/commissioning checks, parameters, results, issues and customer acceptance.
- Operator/maintenance training and attendance; handover manuals, drawings, certificates, backups, spares and signed records; warranty dates and project closure checklist.

Reports · Upcoming FAT/SAT, open punch items, dispatch readiness, site schedule, commissioning, handover completeness and delayed acceptance.

Acceptance · Dispatch reflects FAT/quality/document/commercial controls; sign-offs link to project/machine; blocking punch items prevent configured gate closure; closure requires approved handover and accountable warranty-item transfer.

Sequence / boundary · Uses R2 commercial commitments and R7 readiness. Handover supplies the installed-base/warranty inputs for R9. Spec §13, p.15.

RELEASE SCOPE & ACCEPTANCE

R9

Finance, Workforce & After-Sales

Project cost, cash, people and installed-machine service history are connected.

- AR/AP registers, customer/supplier invoices, due dates, payment allocation and partial payments; milestone invoice requests and project payment visibility.
- Cash, bank/reconciliation, account heads, journals, GL, assets and financial statements where full finance scope is approved.
- Project material, purchase, labour, production, rework, travel and overhead costs; budget versus actual and margin.
- Employee, designation/manager, attendance, leave, overtime, timesheets and project allocation; payroll for approved localization or secure provider integration/export.
- Installed machines, serial/configuration/site, warranty; breakdown/service work, parts/time/travel and sign-off; preventive maintenance, due alerts, spares, retrofits and upgrades.

Reports · Cash/AR/AP and ageing, approved financial statements, project profitability, workforce utilization, attendance/leave/overtime, installed base, warranty, service SLA and maintenance due.

Acceptance · Milestones generate invoice requests; payment status feeds projects; costs reconcile to source transactions; finance/payroll data is restricted; delivered serialised machines have warranty/service timelines.

Sequence / boundary · Uses source transactions from prior releases. Finance depth and payroll countries/localization require explicit scope decisions. Spec §14, p.16.

R10

Enterprise Integration & Intelligence

The platform gains enterprise workflows, integration and cross-module intelligence.

- Configurable workflow/approval matrix by role, user, department, amount, organization and document type; approve/reject/return, delegation, escalation and history.
- Email, dashboard and mobile/web push; optional approved WhatsApp; cross-module dashboards, drill-down, saved filters, exports and scheduled distribution.
- APIs/webhooks, monitoring and controlled accounting, payroll, CAD/BOM and email integrations; approved customer/supplier portals.
- AI-assisted delay, shortage, cost and risk recommendations with source evidence and user confirmation.
- Reconciliation, migration, recovery, performance, security and observability hardening; end-to-end, role, integration, recovery and production-cutover acceptance.

Reports · Executive portfolio, project health, delivery, margin, procurement risk, quality, supplier performance, utilization, service and audit.

Acceptance · Configured approvals enforce authority/sequence; dashboard indicators drill to sources; failed integrations/notifications are visible and safely retryable; restore, performance, security and end-to-end acceptance pass.

Sequence / boundary · Extends earlier module integration and controls. Basic approvals, permissions, audit, validation and backup apply from each module's launch. Spec §15, p.17.

CUSTOMER REQUIREMENT PRESERVATION

Original scope, revised order

Dashboard.docx remains the customer source. Its substantive requirements are retained below; the revised specification reorganizes delivery around the machine-project lifecycle.

Dashboard.docx requirement	Revised delivery
Company, branches, locations, departments; users, roles, permissions and credentials	R1; employee identity in R1, workforce depth in R9
Customer, supplier, item, UOM, cost and selling-price masters	R1; selectable consistently by downstream modules
Enquiry → quotation → sales order → delivery note → invoice → receipt	R2 commercial initiation; R8 delivery; R9 invoicing, receipts and partial payments
Purchase request → RFQ → supplier quote → PO → GRN → supplier invoice → payment	R6 purchasing, GRN, invoice matching and AP handoff; R9 payment processing
Warehouse, stock in/out, transfer, adjustment, cycle count and balance	Warehouse master in R1; transactional inventory in R5
BOM → production planning → work order → material issue → assembly → finished goods	R4 controlled BOM/material planning; R7 production and output
Incoming, in-process and final inspections; NCR and CAPA	R7; FAT/SAT and customer acceptance expand in R8
Employee, attendance, leave, overtime and payroll	R1 identity; R9 workforce/payroll, subject to localization
AR, AP, GL, cash, bank and assets	R9; full statutory finance only where approved
Relational database; master, transaction and reference tables	R1 foundation, extended each release; PostgreSQL remains the repository's accepted system of record
Workflow master, steps/history, approval levels and document types	R1 reference foundation; basic module approvals at launch; advanced configurable workflow in R10
Email, dashboard, mobile and WhatsApp notifications	Module dashboard alerts progressively; enterprise/provider-dependent channels in R10; WhatsApp subject to approval
Sales by customer/month; pending orders	R2; reporting behavior also governed by specification §18
Supplier performance and open PO; stock valuation, ABC and slow-moving	R6 purchase reports; R5 inventory reports
Work-order status, material consumption and productivity	R7 production reports
Trial balance, profit and loss, balance sheet	R9 where full finance is enabled
PO approval, minimum-stock, maintenance-due and leave-request alerts	R6 purchase, R5 stock, R9 service/workforce; channels extended in R10

The original suggested build sequence is superseded by the revised specification's R1–R10 order. "Plain Text" and "Show more lines" are document artefacts; "3GL" means GL. Appendix A's "R18" report references are interpreted as **Section 18**, not an additional release.

DELIVERY GOVERNANCE

Controls across every release

Integration is progressive.

Each module connects to existing source records when introduced. Security, validation, basic approvals, auditability, backup and recovery begin with the foundation and expand continuously.

Shared delivery rules

- Preserve project/machine and source-document references. Transactions are entered once and referenced downstream; no free-text matching for operational traceability.
- Enforce server-side role and record access, controlled status transitions, active master references, numbering, precision/rounding and source quantity/value limits.
- Approved documents change through revisions/amendments. Cancellation records actor, time, reason and downstream impact; posted records use linked reversals or controlled closure.
- Provide permission-controlled attachments, comments/activity, print/export, role-aware drill-down reports and deduplicated actionable notifications.
- Apply transactional posting, database constraints, concurrency and retry/idempotency controls; maintain append-only audit and visible integration failures.

Acceptance gate for each release

Gate	Required evidence
Scope & function	Stories match the specification; happy paths, exceptions, partial processing, permissions and status transitions pass.
Data & quality	Validation, migrations/imports, constraints, audit and report totals reconcile; automated tests and manual regression pass; no critical defect remains.
Security & operations	Sensitive-data/access checks pass; findings are resolved or formally accepted; monitoring, deployment, backup/restore and rollback are documented and tested as applicable.
Business acceptance	Demo completed, evidence retained and authorized business owner signs off. Performance/load and recovery targets are agreed before production acceptance.

Boundary clarifications to retain in implementation

R2 payment milestones describe agreed schedules, not payment receipts. R2 order delivery states must be backed by controlled fulfillment introduced in R8. Clarify order “Approved” versus “Confirmed” in the detailed workflow. R4 material planning connects to live R5 stock and R6 orders progressively. Finance depth, payroll localization, tax rules, CAD formats, notification providers and deployment/recovery objectives follow the specification’s explicit decisions.

Document control

Original customer source: [docs/specs/Dashboard.docx](#)

SHA-256: 3acddd37fca8a16622147664f8cfe52ee6a5a66c5877bc70c867b196577a8df2

Revised scope source: [MachineIQ_ERP_Product_Specification_v1.0.pdf](#)

SHA-256: 2b21f1ce379f43aa234fed79d3556d40e67e5014899692660dd7afec14b40c80

This plan follows the [two-source baseline and decision register](#). Formal specification sign-off and unresolved business rules remain separate from document alignment. Detailed R2 work is in the [R2 implementation plan](#).